



सत्यमेव जयते

NITI Aayog



SECTOR AT A GLANCE 01

Urban: Waste Management

WASH in India

DRAFT

SECTOR AT A GLANCE 02

URBAN: WASTE MANAGEMENT

WASH IN INDIA

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). Urban – Waste Management. NITI Aayog Sector-at-a-Glance Series – WASH in India, 2023-WASH-SAG-02 <link>

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Overview

Sanitation infrastructure and wastewater treatment capacity have witnessed a growing focus in urban India since the launch of the Swachh Bharat Mission (Urban) or SBM-U in 2014. SBM-U 2.0, launched in October 2021, emphasises the safe management of solid and liquid waste at scale to attain “garbage-free cities”, and fully realise health benefits from sanitation investments, as well as attain environmental health outcomes.¹ Furthermore, effective waste management is critical to India’s circular economy efforts and can drive greater climate resilience, economic productivity, and a range of sustainability outcomes.²

However, despite this progress and focus, significant gaps remain in urban waste management such as the lack of waste segregation at source and waste collection, alongside insufficient waste treatment capacity in Indian cities and limited exploration and adoption of new waste technologies.

Within urban waste management, solid waste management, faecal waste management, used water management, capacity building, and skill development have been major policy focus areas recently. Several ongoing government programs within the Swachh Bharat Mission-Urban 2.0 focus on sustainable urban waste management. These efforts, along with several state- and local-level policies and initiatives, seek to ensure that urban India is clean, open-defecation-free, and environmentally sustainable.

This note covers key trends in urban waste management and lists key opportunities and challenges, as well as ideas for the way forward.

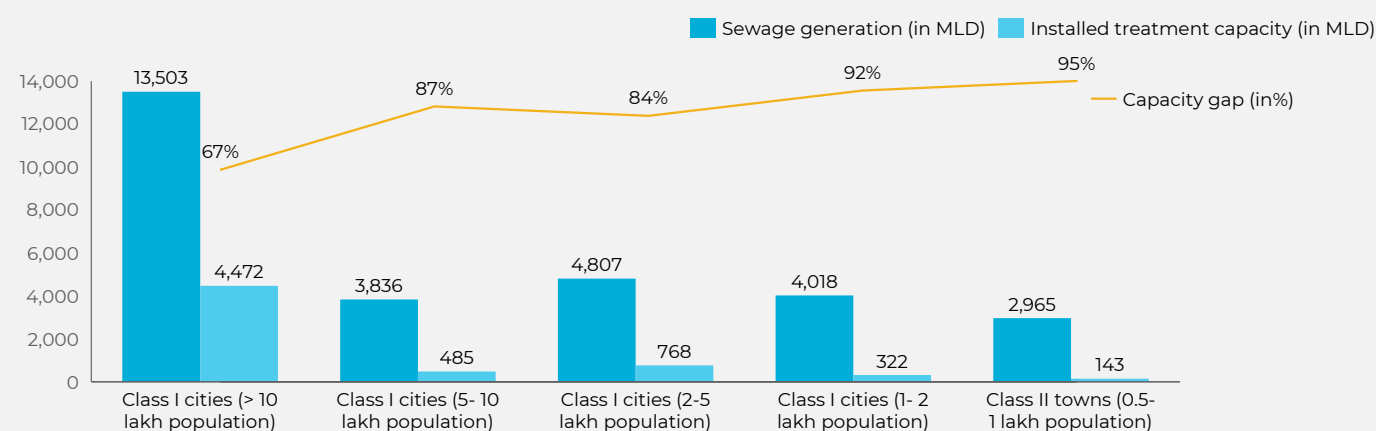
Figure 1: Key urban waste management statistics



Source: Waste processing capacity- NITI Aayog, Waste Wise Cities- Best practices in municipal solid waste management, 2021; Toilets relying on OSS- NITI Aayog, Faecal sludge and septage management in urban areas, 2021; Sewage generation- NITI Aayog, Urban Wastewater Scenario in India, 2022; Ministry of Jal Shakti, Sewage water treatment, 2022

Note: * MLD: Million litres per day

Figure 2: Wastewater generation and treatment capacity gap in Class I cities and Class II towns (2022)



Source: NITI Aayog, Urban Wastewater Scenario in India, 2022

1 MoHUA, Municipal solid waste management Manual, 2016; NITI Aayog, Waste Wise Cities, 2021
2 World Bank, Towards a circular economy: Addressing the waste management threat, 2022

Key trends

This section dives deeper into India's urban waste management landscape, outlining statistics and trends on:

1. Solid waste management
2. Faecal waste management (FSM)
3. Used water management
4. Capacity building and skill development

Solid waste management

Solid waste management includes the effective segregation, collection, transportation, disposal, recycling, and treatment of solid waste, as well as the prevention of littering.

India's waste management policy has undergone a significant shift:

The last few years have witnessed a growing focus on waste segregation and door-to-door collection in India and a significant increase in the country's waste processing capacity from 18% in 2014 to 70% in 2021.³ Guidelines for the use of waste, such as the use of plastic waste in cement kilns, have also been issued.⁴

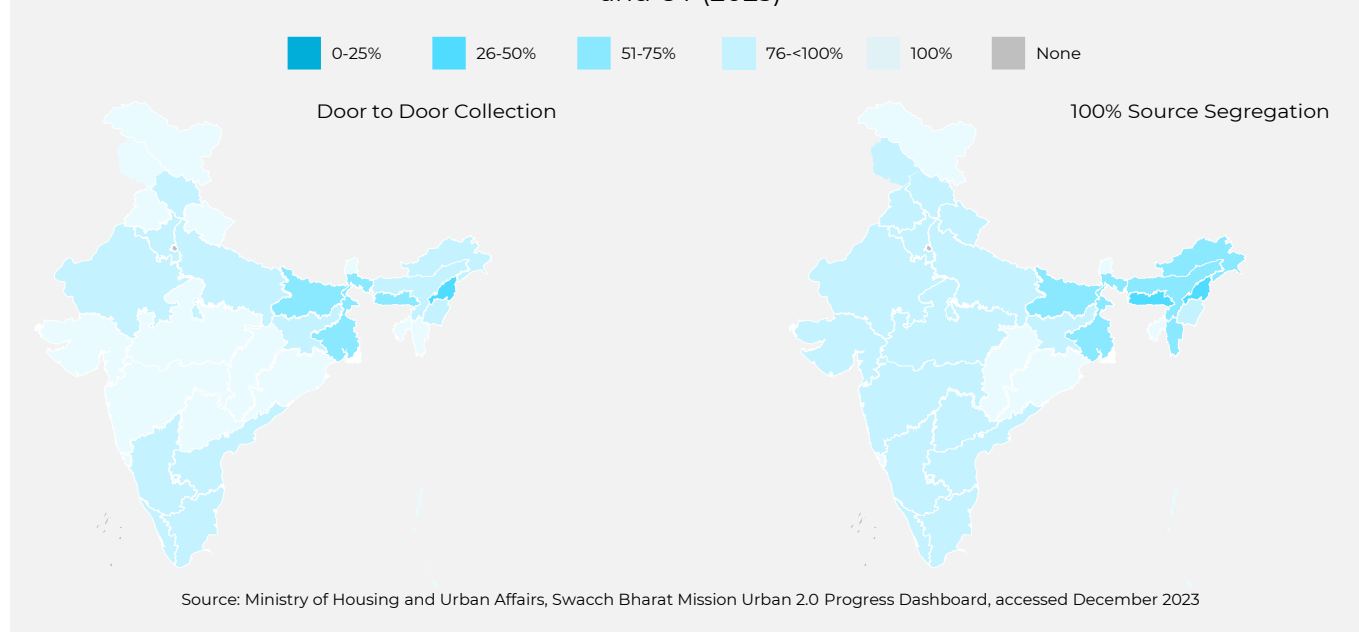
Responsibility of plastic producers for recycling attracts increased emphasis:

The Plastic Waste Management Rules of 2016 and its amendments thereafter focus on the disposal and recycling of single-use plastics.⁵ The extended producer responsibility and buyback model places the onus on plastic producers to pay for reuse and recycling.⁶

Community-based solutions have been vital in addressing open dumping and littering:

Women SHGs have undertaken door-to-door collection systems of solid waste from houses and commercial establishments in some towns.⁷ This was accompanied by mobilization, training, monitoring, and healthcare initiatives.⁸

Figure 3: Percentage of wards with 100% door-to-door collection and source segregation by state and UT (2023)



3 NITI Aayog, Waste Wise Cities, 2021

4 CPCB, Guidelines for Co-processing of Plastic Waste in Cement Kilns, 2017

5 NITI Aayog, Sustainable urban plastic waste management, 2021

6 NITI Aayog, Sustainable urban plastic waste management, 2021

7 MoHUA, Critical role of community-based organizations in urban sanitation and waste management, 2019

8 MoHUA, Critical role of community-based organizations in urban sanitation and waste management, 2019

Faecal waste management

Faecal waste management encompasses programs to safely store, transport, treat, and then extract value from human waste with the overall objective of enhancing human health and environmental sustainability.

Common standards for treatment infrastructure across urban India:

The Ministry of Housing and Urban Affairs (MoHUA) has played an active role in setting guidelines and developing solutions for FSM in urban India, supported by National Green Tribunal orders to enable effective sludge management.⁹ This has allowed ULBs and municipalities to draw from common standards and best practices while awarding contracts and tenders for faecal sludge management and related sanitation infrastructure.¹⁰

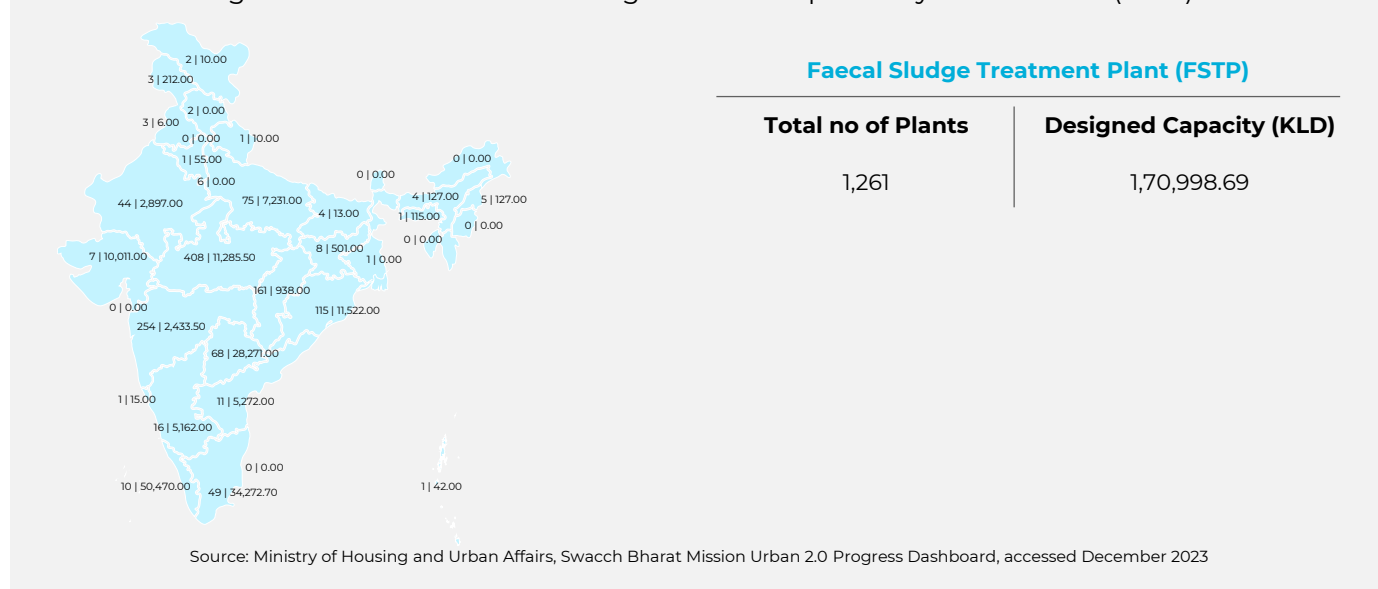
Policy and financing support has accelerated FSTP setup and coverage:

The National Policy on Faecal Sludge and Septage Management 2017 provided the impetus for the uptake of FSM projects by state governments in India, from a single operational faecal sludge treatment plant (FSTP) in 2015 (in Devanahalli) to over 1200 operational FSTPs currently.¹¹

Increased investments for setting up STPs and FSTPs and adopting a hybrid approach:

Atal Mission for Rejuvenation and Urban Transformation 2.0 (AMRUT 2.0) and SBM-U 2.0 place a greater emphasis on wastewater treatment, with central, state, and local governments sharing responsibility for STP and FSM expenditure. The complementary role played by centralized STPs and more decentralized FSTPs has now been recognized as an important approach to ensuring citywide waste treatment coverage in a financially viable manner.¹²

Figure 4: Number of faecal sludge treatment plants by state and UT (2023)



Used water management

Used water management involves the collection, treatment, and reuse of mildly polluted water from domestic and institutional sources. A range of self-managed, community-managed, public-sector, and private-sector models are used in the management of used water.

⁹ MoHUA, Circular Economy in Municipal Solid and Liquid Waste, 2021

¹⁰ Ministry of Jal Shakti, Manual: Faecal sludge management, 2021

¹¹ NITI Aayog, Faecal sludge and septage management in urban areas, 2021; CSE, The Policy & Practice Forum 2023, 2023

¹² MoHUA, SBM (U) 2.0 Operational guidelines, 2021

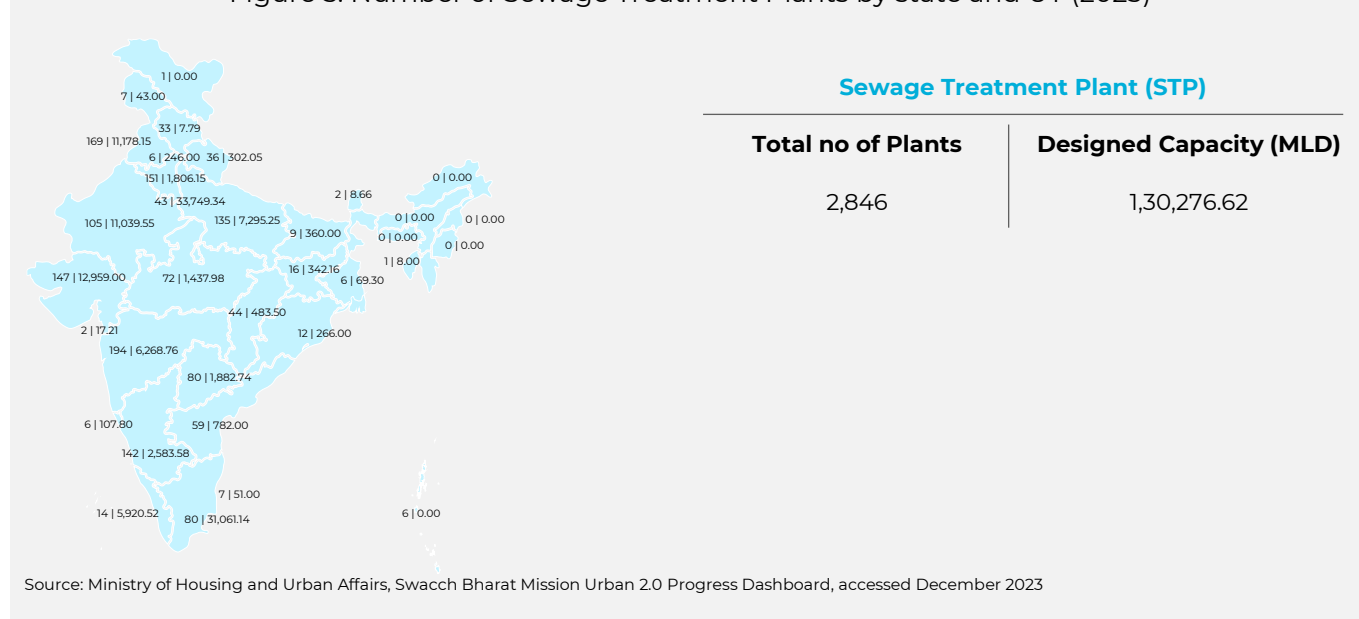
Increase in greywater and used water management across households, communities, and institutions:

Measures have been undertaken to improve water availability by recycling greywater from sinks, bathrooms, and kitchens across households, factories, and hospitals.¹³ This includes treatment through disinfection, filtration, etc.¹⁴

Used water management requires significant behaviour change:

Used water management requires significant shifts in behaviour both in terms of segregating water flows to ensure that used water does not mix with more contaminated water and on the demand side to ensure that treated used water gets used by individuals, communities, and institutions for purposes that don't require fully treated water.

Figure 5: Number of Sewage Treatment Plants by state and UT (2023)



Capacity building and skill development

This pillar includes community engagement and training programs for citizens, sanitation workers, and solid and liquid waste management personnel, at a local, state, and central level, to enable them to more effectively manage waste management infrastructure as well as carry out essential processes and enact key behaviours.

National capacity building framework to enable reaching a “solid waste-free” status:

Conceptualized under SBM-U 2.0, a national framework to enable India to reach solid waste-free status has been created and it focuses on capacity strengthening of State & municipal cadres, capacity augmentation through specialized HR, and overall ecosystem strengthening.¹⁵

Mechanised Sanitation Ecosystem scheme deployed to ensure the safety and dignity of sanitation workers:

The National Action for Mechanised Sanitation Ecosystem scheme is a scheme by the Ministry of Social Justice and Empowerment (MoSJE) as a joint initiative of the MoSJE and the MoHUA.¹⁶ It ensures the safety and dignity of sanitation workers by creating an enabling environment that recognizes them as a key contributor to the sanitation economy and provides them access to mechanized equipment, safety equipment, and related training.¹⁷

¹³ Ministry of urban development, Water efficiency and conservation in Urban India, 2017

¹⁴ Ministry of urban development, Water efficiency and conservation in Urban India, 2017

¹⁵ MoHUA, National capacity building framework for garbage-free cities, 2022

¹⁶ Ministry of Social Justice and Empowerment, NAMASTE, 2022

¹⁷ Ministry of Social Justice and Empowerment, NAMASTE, 2022

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective urban waste management practices. These revolve primarily around the potential for improved health, reduced emissions, improved economic productivity and social equity.



Reduction in GHG emissions:

In India, waste-related greenhouse gas (GHG) emissions are estimated at 0.7 million tonnes of CO₂ equivalent, primarily from methane emissions in landfills and wastewater.¹⁸ Maximising resource recovery from waste, both material and energy, along with proper disposal, ensures effective waste management and contributes to significant GHG emission reductions.¹⁹



Improvement in economic productivity:

Improving the infrastructure for sewage treatment can help reduce economic losses caused by inadequate sanitation.²⁰



Improvement in social equity and economic outcomes for sanitation workers:

Improved urban waste management empowers informal waste pickers and sanitation workers, recognising the economic value of their work. Mechanising cleaning and providing safety measures for sanitation workers further improve their livelihoods and living standards. This enhances social equity and economic outcomes, benefiting waste workers and the urban environment.²¹

Challenges

This section highlights some of the major challenges and constraints facing urban waste management practices in India. Efforts to resolve such challenges will ensure higher waste collection, improved treatment capacity, better citizen behaviours, and overall system efficiency.



Limited waste segregation with gaps in treatment capacity and efficiency:

As per a 2021 report, only ~50% of solid waste generated in India was treated.²² Adding to this, about 11% of wards lack segregation of waste at source, and about 5% of wards still lack door-to-door waste collection.²³ Industrial solid waste disposal is also inadequate due to a lack of requisite infrastructure and negligence by industries in following safeguards.²⁴ Similarly, only 40% of wastewater in urban India is safely treated, with over 60% left untreated and entering water bodies.²⁵ Despite the setup of several FSTPs and STPs over the last few years, there is still a large unmet demand for the safe and efficient treatment of wastewater generated from onsite and offsite sanitation systems.²⁶



Limited awareness amongst the general public:

Many communities, cities, and towns continue to dispose of waste through dumping, open burning, or depositing it in low-lying areas outside the city, which needs immediate action.²⁷

18 TERI, Managing India's Waste Key To Energy, Climate Benefits, 2019

19 Indian Institute of Public Administration, Climate Smart Governance, 2020

20 World Bank, Wastewater A Resource that Can Pay Dividends for People, the Environment, and Economies, Says World Bank, 2020

21 ORF, Solid Waste Management in Urban India: Imperatives for Improvement, 2020

22 CPCB, Annual Report on Solid Waste Management (2020-21), 2021

23 MoHUA, Swachh Bharat Mission, 2022

24 CPHEEO, Industrial Solid Waste, accessed September 2023

25 WHO UNICEF, WASH Data, 2020

26 WHO UNICEF, WASH Data, 2020

27 ORF, Solid Waste Management in Urban India: Imperatives for Improvement, 2020

Such practices can result in contamination of water sources, air pollution, and soil degradation.



Limited formalisation and inefficiencies within e-waste and construction and demolition waste management:

India's e-waste collection, transportation, processing, and recycling value chain is dominated by the informal sector and remains mostly unregulated.²⁸ However, the value recovery from collected e-waste has the potential to be further enhanced.²⁹ The construction industry in India generates about 150-500 million tonnes of C&D waste every year. This brings many challenges to the forefront such as unauthorised dumping, a lack of space for disposal, and improper mixing with biodegradable waste.³⁰



Underdeveloped waste treatment markets and usage of old technology:

The lack of demand to segregate and treat greywater has led to limited enterprise presence within the sector. Capital requirements for research, and development as well as setup costs have also acted as barriers to entry.³¹ Further, India is heavily dependent on older technologies for handling waste management.³² Limited funds and high capex requirements lead to the adoption of older technologies with lower capital costs despite poor performance benchmarks.³³



Inadequate availability of trained professionals in sanitation services:

For a sustainable sanitation value chain, an efficient delivery system that comprises an adequate number of trained planners, sanitary inspectors, and engineers is required to plan, implement, and monitor city-level sanitation programs.³⁴ As per data, there is a significant gap of around 300,000 certified plumbers apart from mechanical engineers, statisticians, and other technical roles.³⁵

28 TERI, E-Waste Management in India: Challenges and Opportunities, 2019

29 TERI, E-Waste Management in India: Challenges and Opportunities, 2019

30 MoHUA, National workshop on "Recent development with recycling and use of C&D waste in construction sector", 2024

31 Ministry of Jal Shakti, Greywater management: Brochure, accessed July 2023

32 NITI Aayog, Urban Wastewater Scenario in India, 2022

33 NITI Aayog, Urban Wastewater Scenario in India, 2022

34 Indian Sanitation Coalition, Skilling in Sanitation, 2017

35 Indian Sanitation Coalition, Skilling in Sanitation, 2017

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee urban waste management and related practices.



Ministry of Housing and Urban Affairs

The Ministry is concerned with matters related to urban development and aims to tackle the various challenges that arise in the wake of rapid urbanisation in India. This includes themes around housing, poverty, unemployment, the development of health, sanitation, and education infrastructure, and so on.³⁶ Thus overseeing and implementing policies around effective urban waste management also falls under its ambit. The ministry is responsible for the implementation of the Swachh Bharat Mission (Urban) 2.0 with the aim of making Indian cities “garbage-free” in collaboration with states/UTs and ULBs.³⁷



Ministry of Environment, Forests, and Climate Change

Promotes the safe management and use of hazardous substances and waste; sets and notifies rules and guidelines for waste management and provides financial assistance to states to run awareness programmes.³⁸



Central Pollution Control Board³⁹

The Board is responsible for the prevention, control, and abatement of water pollution (thus including pollution caused by industries) in accordance with the Water (Prevention and Control of Pollution) Act, 1974. Under its functions, the Board looks into matters related to urban pollution, industrial pollution, and waste management including e-waste, municipal solid waste, plastic waste, and other related activities.

Major policies and programs

This section highlights existing policies related to urban waste management and their key features. A more comprehensive list of policies is provided in the policy repository of the NITI for States platform.



Swachh Bharat Mission-Urban 2.0⁴⁰

The SBM-U entails the following key measures to achieve sustainable waste management in urban areas:

- Promoting sustainable solid waste management with a focus on phasing out single-use plastic, source segregation and scientific management of all fractions of waste
- Promoting sustainable sanitation and treatment of used water by providing end-to-end solutions
- Provisioning for treatment for wastewater before discharge
- Eliminating manual scavenging
- Promoting behaviour change about waste segregation and the three Rs—reduce, recycle and reuse
- Building capacities of ULBs and sanitation workforce

This program also houses the Swachh Survekshan 2023 program which aims to create garbage-free cities by following the philosophy of ‘Waste to Wealth’.

³⁶ Lok Sabha Secretariat, Background Note on “Housing and Urban Affairs”, 2021

³⁷ MoHUA, Swachh Bharat Mission - Urban Guidelines, 2021

³⁸ Ministry of Environment, Forest and Climate Change, About, accessed December 2023

³⁹ Ministry of Jal Shakti, About the Ministry, accessed in December 2023

⁴⁰ MoHUA, Swachh Bharat Mission - Urban Guidelines, 2021

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector.

The urban waste management sector stands at a pivotal point in India. On the one hand, government programs and policies have recognised the importance of comprehensive waste management and attaining solid waste-free cities and have also recognised the importance of managing different waste streams, while protecting the livelihoods of sanitation workers and ensuring their safety. There have also been very exciting developments in decentralized waste management with hundreds of FSTPs being implemented by cities and States across the country.

On the other hand, however, several challenges will need to be addressed to ensure the sustainability of these programs and achieve true progress. Many of the business models, especially for FSTPs, are still not financially viable, sanitation workers continue to face extreme health hazards, and e-waste, and plastics-waste are still being managed primarily through the informal sector.

Some ideas that could spur progress include:

 Fostering innovative financing for urban waste management models	• Encourage state governments to partner with commercial banks, philanthropists, development banks, and impact investors to (i) Create blended finance instruments and viable financing models for urban waste management projects (ii) ensure long-term financial sustainability and attract diverse stakeholders.
 Support data-driven decision-making for waste management	• Invest in collecting detailed data on both unit economics of different waste treatment models and usage rates across the value chain to enable Urban Local Bodies (ULBs) to develop better tenders, attracting more private sector players. • Utilise innovative data sources such as satellite imagery to develop tools and platforms for ULBs to predict ideal locations for waste treatment facilities, estimate usage rates, and assess financial viability, leading to more efficient waste planning.
 Increasing enforcement of Extended Producer Responsibility (EPR)	• Strictly enforce Extended Producer Responsibility (EPR) measures, particularly for plastic and e-waste to (i) hold companies accountable for managing their waste (ii) incentivise them to design more sustainable products.



**Investing in
behaviour
change
programs**

- Invest in and implement behaviour science-based programs aimed at shifting citizen behaviour, especially promoting source segregation of waste which will be crucial for achieving waste-free city goals.

DRAFT

Recommended readings

NITI Aayog. (2021). *Waste Wise Cities- Best practices in municipal solid waste management*
<https://www.niti.gov.in/sites/default/files/2021-12/Waste-Wise-Cities.pdf>

NITI Aayog. (2021). *Faecal sludge and septage management in urban areas*
<https://www.niti.gov.in/sites/default/files/2021-08/NITI-NFSSM-Alliance-Report-for-digital.pdf>

NITI Aayog. (2022). *Urban Wastewater Scenario in India*
https://www.niti.gov.in/sites/default/files/2022-09/Waste-Water-A4_20092022.pdf

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